

PROJECT DESIGN PHASE

Problem–Solution Fit

Project Title

Optimization of Users, Groups, Roles with Access Control and Workflows in ServiceNow

Introduction

Organizations often struggle to manage users, groups, and roles manually. As the number of employees increases, maintaining secure access becomes more difficult. Manual processes result in delays, incorrect permissions, duplicate records, and increased security risks.

The proposed ServiceNow solution addresses these problems by automating user management, implementing role-based access control (RBAC), configuring Access Control Lists (ACLs), and using Flow Designer to automate approval workflows.

Problems Identified

- Manual user creation and maintenance.
- Incorrect role assignments.
- Unauthorized access to sensitive data.
- Time-consuming approval process.
- Lack of centralized access management.
- Human errors during administration.
- Difficulty in auditing user permissions.

Proposed Solution

The proposed ServiceNow application provides:

- Centralized user management.
- User group organization.
- Role-based authorization.
- Secure Access Control Lists (ACLs).
- Automated approval workflows.
- Email notifications.
- Improved auditing and reporting.

Problem–Solution Mapping

Problem| Proposed Solution

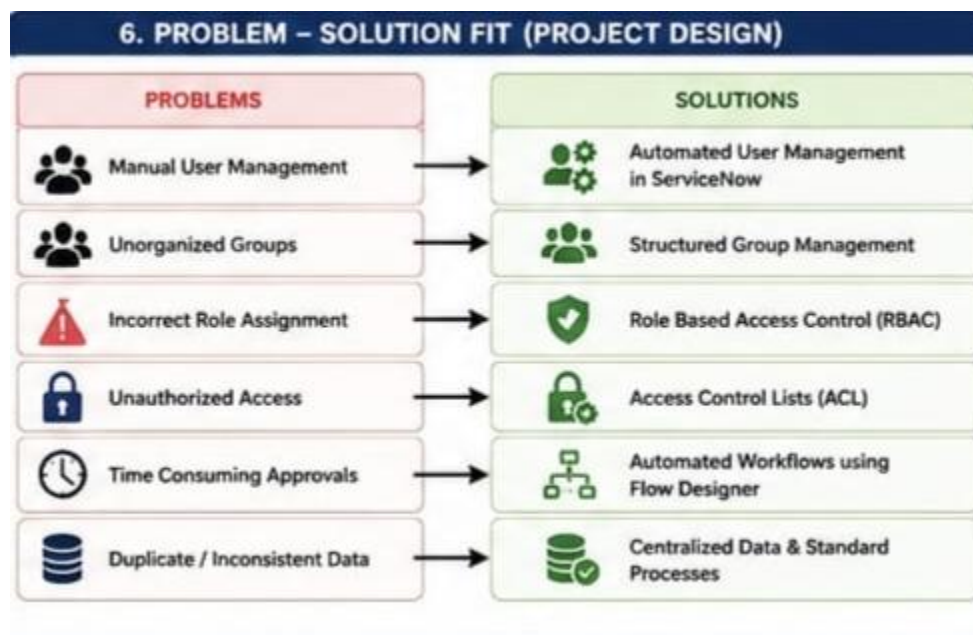
Manual user management| Automated user management in ServiceNow
Incorrect role assignment| Role-Based Access Control (RBAC)
Unauthorized access| Access Control Lists (ACLs)
Slow approvals| Flow Designer automation
Duplicate records| Centralized user management
Security issues| Controlled user permissions

Expected Benefits

- Improved security.
- Faster approvals.
- Reduced administrative effort.
- Better compliance.
- Increased productivity.

Conclusion

The proposed solution successfully addresses the identified business challenges by integrating secure access control, workflow automation, and centralized administration within the ServiceNow platform.



Proposed Solution

Overview

The proposed solution is designed to automate and optimize user, group, and role management using the ServiceNow platform. The solution ensures that only authorized users can access specific modules by implementing Role-Based Access Control (RBAC) and Access Control Lists (ACLs).

Workflow automation using Flow Designer minimizes manual intervention and speeds up approval processes.

Solution Components

User Management

- Create user accounts.
- Maintain employee details.
- Update user information.

Group Management

- Create organizational groups.
- Add users to groups.
- Manage department-wise users.

Role Management

- Create roles.
- Assign roles to users.
- Manage user permissions.

Access Control

- Configure ACL rules.
- Restrict unauthorized access.
- Protect sensitive records.

Workflow Automation

- Automate approval requests.
- Send notifications.
- Track request status.

Advantages

- Secure system.
- Reduced manual effort.
- Centralized administration.

- Better audit tracking.
- Faster processing.

Conclusion

The proposed solution integrates all access management activities into a single ServiceNow platform, improving operational efficiency and maintaining organizational security.

Solution Architecture

The architecture consists of the following components:

Users



ServiceNow Login



User Management



Group Management



Role Assignment



Access Control Lists (ACLs)



Flow Designer Workflow



Approval Process



Notifications



Authorized User Access

Component Description

User Layer

Users log in to the ServiceNow platform and submit access requests.

User Management

Maintains employee information and user accounts.

Group Management

Organizes users into appropriate organizational groups.

Role Management

Assigns permissions according to organizational responsibilities.

Access Control Lists

Validates whether users are authorized to access records or modules.

Workflow Layer

Automates approval requests and notification processes.

Database

Stores users, groups, roles, approvals, and audit records.

Benefits

- Secure architecture.
- Centralized management.
- Automated approvals.
- Scalable solution.
- Easy maintenance.

Conclusion

The proposed architecture ensures secure, efficient, and automated access management while maintaining data integrity and organizational compliance.

Architecture Diagram

